Information Theory Workshop (ITW) 2026

Information-theoretic foundations of anytime-valid inference

Part 2 (11:00 am to 12:30pm)

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Tutorial outline

PART 1:

Introduction + finite alphabet methods

1. Motivation

A/B testing example; failure of classical methods; Anytime-valid (AV) testing

2. Fundamental Limits

Classical DPI + Wald arguments

3. AV Test for Finite Alphabets

Universal Inference; Proof of optimality

PART 2:

General Principles

1. Bounded mean testing

Testing-by-betting principle, Kelly criterion, universal portfolios.

2. Unifying Design Principle

Donsker-Varadhan representation, Likelihood-free Universal Inference.

3. Extensions

Sequential estimation, change detection, non-iid data, etc.

Recap: the sequential testing problem

- Sequential testing problem on $\mathcal{X}$:

$$X_1, X_2, \ldots \overset{\text{iid}}{\sim} P.$$

$$H_0 : P_X \in \mathcal{P}_0 \qquad \text{vs.} \qquad H_1 : P_X \in \mathcal{P}_1.$$

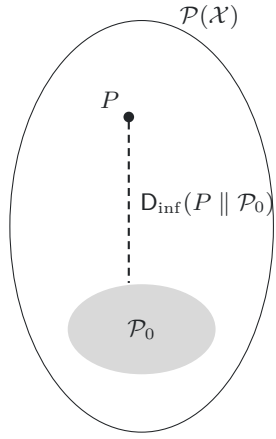
- Design a stopping time τ_α at which we reject H_0, with

$$\underbrace{\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau_\alpha < \infty) \leq \alpha,}_{\text{Validity under } H_0} \qquad \underbrace{\inf_{P \in \mathcal{P}_1} \mathbb{P}_P(\tau_\alpha < \infty) = 1\,.}_{\text{“Power-one” under } H_1}$$

- Any such τ_α must satisfy

$$\mathbb{E}_P[\tau_\alpha] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)},$$

where $\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) := \inf_{Q \in \mathcal{P}_0} D(P\|Q)$.



Goal: Design τ_α that matches this limit as $\alpha \downarrow 0$.

Recap: universal inference

- In Part 1, we focused on $\mathcal{X} = \{1, \ldots, k\}$.

- **General approach:**
  (i) Define a process $\{E_t : t \geq 1\}$ that quantifies evidence against H_0.
  (ii) Reject H_0 when E_t reaches the threshold $1/\alpha$.

- In Part 1, we used a mixture-to-maximum likelihood ratio process:

$$E_t = \frac{\displaystyle\int_{\Delta_{k-1}} \prod_{i=1}^{t} q(X_i)\, \pi(dq)}{\displaystyle\sup_{Q \in \mathcal{P}_0} \prod_{i=1}^{t} Q(X_i)} = \frac{\text{mixture likelihood}}{\text{maximum null likelihood}}.$$

- However, in many problems, a full likelihood is unavailable or difficult to compute.

A nonparametric example: bounded-mean testing

- **Observations:** $\mathcal{X} = [0,1]$, and $X_1, X_2, \ldots \overset{\text{iid}}{\sim} P$
- **Testing problem:** For some fixed $m \in (0,1)$, decide b/w

$$H_0 : \mathbb{E}_P[X] \leq m, \quad \text{versus} \quad H_0 : \mathbb{E}_P[X] > m.$$

- The null posits that the mean of this stream equals m; equivalently,

$$\mathcal{P}_0 = \big\{P \in \mathcal{P}([0,1]) : \mathbb{E}_P[X] \leq m\big\}.$$

- **There is no common σ-finite dominating measure** for $\mathcal{P}_0$.
- The same is true for many other problems; e.g., *two-sample testing, independence testing etc.*

The principle of testing by betting

Evidence against a null hypothesis can be formally quantified as the gain in wealth of a fictitious bettor repeatedly betting on observed outcomes, at odds that are fair or subfair under H_0.

Initialize: $W_0 \leftarrow 1$.

For $t = 1, 2, \ldots$ **do:**

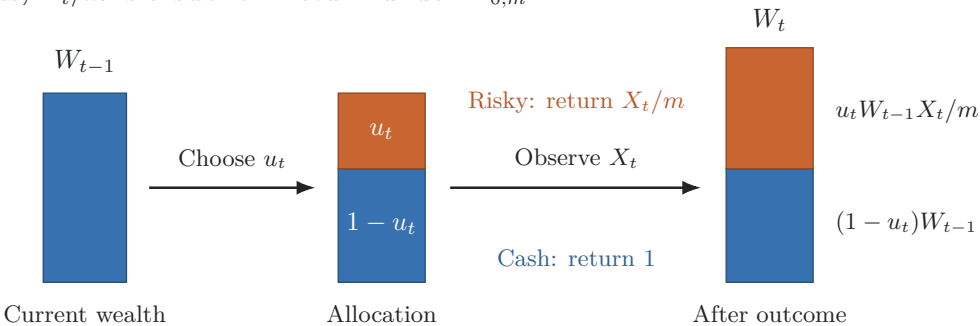
1. **Place a bet:** Choose a fraction $u_t \in [0, 1]$ of current wealth to bet, and the payoff rule.
2. **Observe** the next outcome X_t, and collect the payoff.
3. **Update** the bettor's wealth W_t.

End for.

Designing a fair bet for $H_{0,m} : \mathbb{E}[X_t] \leq m$

$$\mathbb{E}_Q[X_t] \leq m \quad \Longrightarrow \quad \mathbb{E}_Q\left[\frac{X_t}{m}\right] \leq 1, \quad \text{for all } Q \in \mathcal{P}_{0,m}.$$

Thus, X_t/m is a sub-fair return under $H_{0,m}$.



$$W_t = W_{t-1}\left((1-u_t) + u_t\frac{X_t}{m}\right), \qquad 0 \leq u_t \leq 1.$$

Testing by betting for bounded means

Testing $H_{0,m}: \mathbb{E}[X] \leq m$, with $X_t \in [0,1]$.

Initialize: $W_0 = 1$.

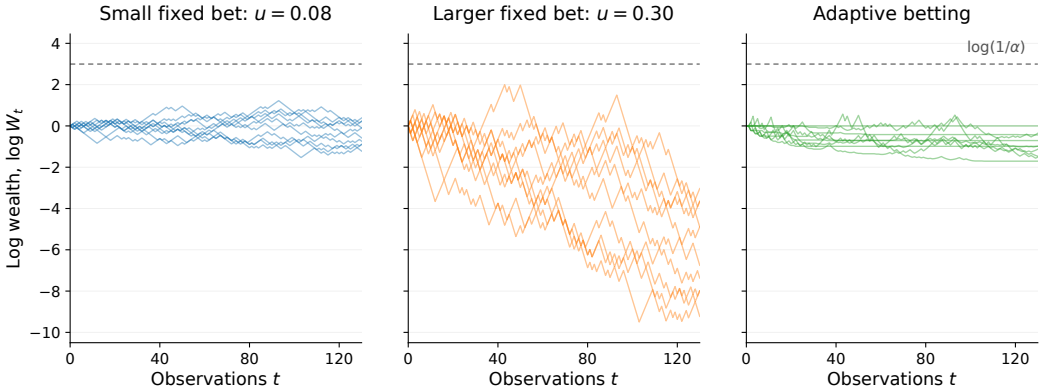
For $t = 1, 2, \ldots$:

1. **Choose** $u_t \in [0,1]$ using only X^{t-1}.
2. **Observe** X_t.
3. **Update** $W_t = W_{t-1}\big(1 - u_t + u_t X_t / m\big)$.
4. **Reject** H_0 and stop if $W_t \geq 1/\alpha$.

How should we choose the sequence of bets $\{u_t\}$?

Testing by betting for bounded means

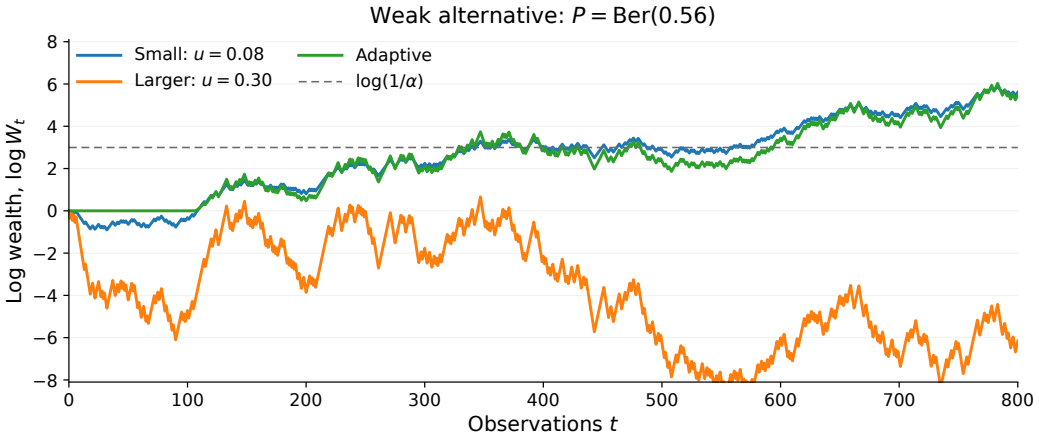
Under H_0: $X_t \sim \text{Ber}(0.50)$. Ten independent runs with the same data for all bets.



Large gains are unlikely under H_0, irrespective of the betting strategy.

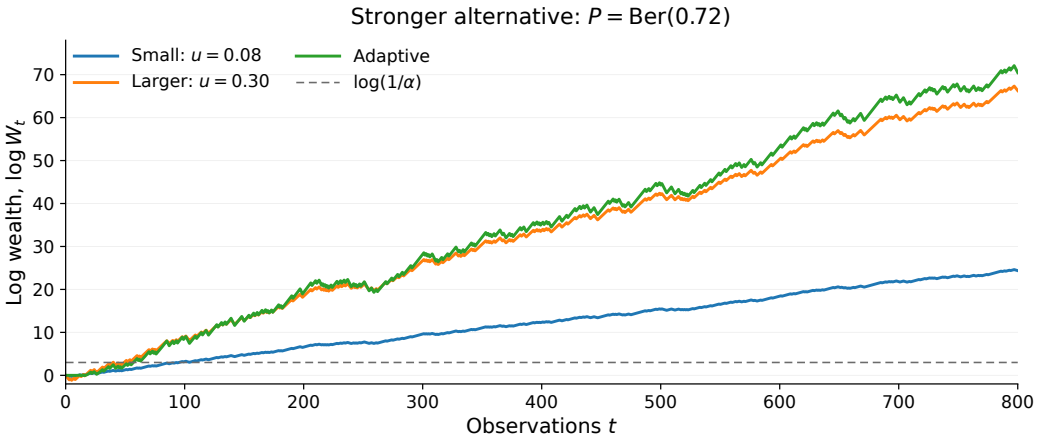
Testing by betting for bounded means

Weak alternative: the smaller bet wins; adaptive stays competitive.



Testing by betting for bounded means

Stronger alternative: the larger fixed bet wins; adaptive stays competitive.



Any betting strategy works under $H_{0,m}$

Recall that $u_t \in [0,1]$ is chosen using only $\mathcal{F}_{t-1} = \sigma(X_1, \ldots, X_{t-1})$, and

$$W_t = W_{t-1}\left(1 - u_t + u_t \frac{X_t}{m}\right), \qquad W_0 = 1.$$

Under any $Q \in \mathcal{P}_0$,

$$\mathbb{E}_Q[W_t \mid \mathcal{F}_{t-1}] = W_{t-1}\left(1 - u_t + u_t \frac{\mathbb{E}_Q[X_t]}{m}\right) \le W_{t-1}.$$

Thus, $\{W_t\}_{t\ge 0}$ is a **nonnegative supermartingale** under H_0.

By Ville's inequality, we have

$$\sup_{Q \in \mathcal{P}_{0,m}} \mathbb{P}_Q\left(\sup_{t\ge 0} W_t \ge \frac{1}{\alpha}\right) \le \alpha, \quad \text{hence} \quad \tau_\alpha = \inf\left\{t \ge 1 : W_t \ge \frac{1}{\alpha}\right\} \text{ is valid for } \mathcal{P}_0$$

Ville. *Étude critique de la notion de collectif* (1939)

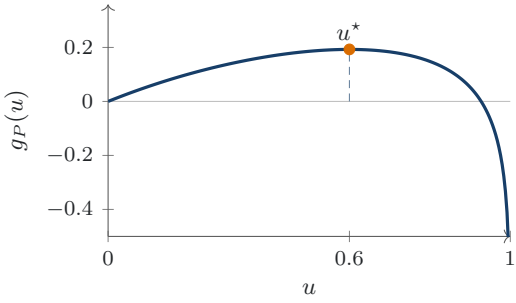
How should we bet? The Kelly criterion

Suppose we use a fixed $u \in [0,1]$ in every round:

$$W_n(u) = \prod_{t=1}^{n} \left(1 - u + u\frac{X_t}{m}\right).$$

Under an i.i.d. alternative P, by the SLLN,

$$\frac{1}{n}\log W_n(u) = \frac{1}{n}\sum_{t=1}^{n} \log\left(1 - u + u\frac{X_t}{m}\right) \xrightarrow{\text{a.s.}} \underbrace{\mathbb{E}_P\left[\log\left(1 - u + u\frac{X}{m}\right)\right]}_{=:g_P(u)}.$$

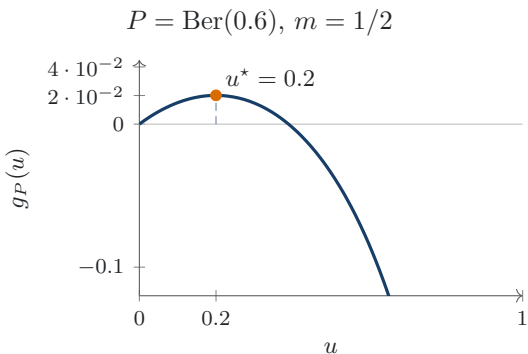


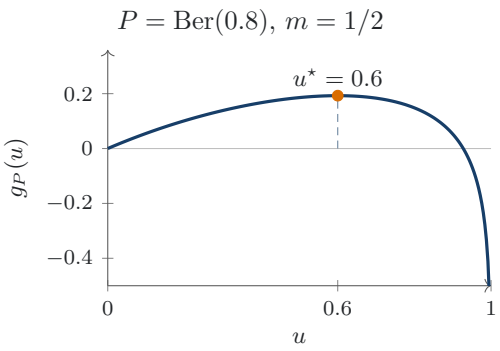
The **log-optimal or Kelly bet** is

$$u^\star \in \arg\max_{u \in [0,1]} g_P(u).$$

Learning the Kelly bet by mixing

Challenge. The Kelly bet $u^\star$ depends on the unknown alternative P.





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Idea. Instead of committing to a single u, mix over all constant allocations $u \in [0,1]$.

$$\overline{W}_t = \int_0^1 W_t(u)\,\pi(du), \qquad W_t(u) = \prod_{i=1}^{t} \left(1 - u + u\frac{X_i}{m}\right).$$

Learning the Kelly bet by mixing

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$$\overline{W}_t = \int_0^1 W_t(u)\,\pi(du), \qquad W_t(u) = \prod_{i=1}^{t} \left(1 - u + u\frac{X_i}{m}\right).$$

Validity. Mixing preserves the nonnegative supermartingale property under every $Q \in \mathcal{P}_0$.

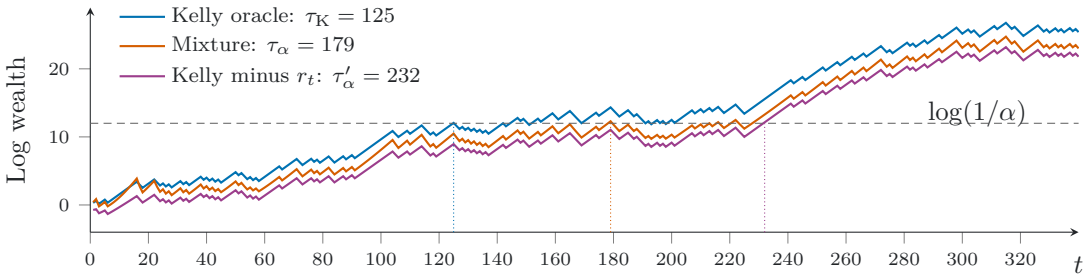
Efficiency? How much log-wealth does the mixture lose relative to the Kelly bet $u^\star$?

From portfolio regret to stopping time

For mixing distribution $\pi = \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$, we have a pathwise guarantee

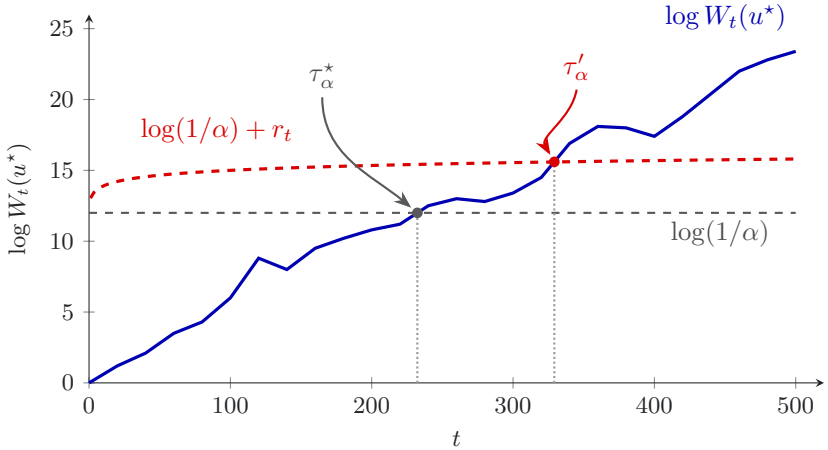
$$\log \overline{W}_t \geq \sup_{u \in [0,1]} \log W_t(u) - r_t, \qquad r_t = \tfrac{1}{2}\log(t+1) + \log 2.$$

Hence, $\tau_\alpha \leq \tau'_\alpha := \inf\{t \geq 1 : \log W_t(u^\star) - r_t \geq \log(1/\alpha)\}$.



From portfolio regret to stopping time

$$\tau'_\alpha = \inf\left\{t \geq 1 : \log W_t(u^\star) \geq \log(1/\alpha) + r_t\right\}, \qquad r_t = \frac{1}{2}\log(t+1) + \log 2.$$



From portfolio regret to stopping time

$$\tau'_\alpha = \inf\{t \geq 1 : \log W_t(u^\star) \geq \log(1/\alpha) + r_t\}, \qquad r_t = \frac{1}{2}\log(t+1) + \log 2.$$

By the regret bound, we have $\tau_\alpha \leq \tau'_\alpha$.

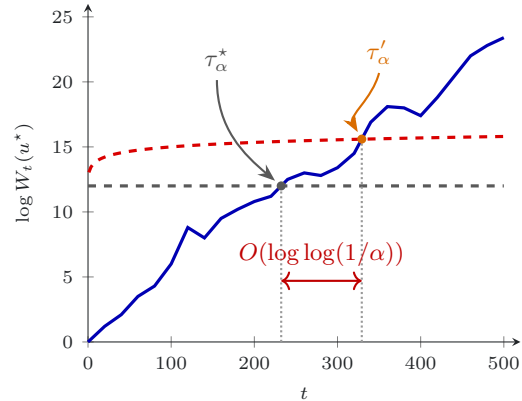
Under P, the oracle process grows linearly:

$$\frac{1}{t}\log W_t(u^\star) \to g_P^\star > 0.$$

Hence, the test τ_α is **power one**.

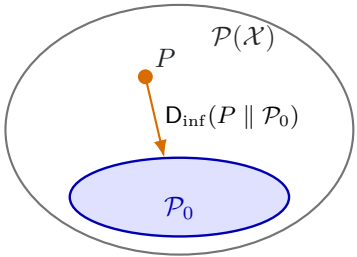
Since $\tau_\alpha \leq \tau'_\alpha$, we also have

$$\mathbb{E}_P[\tau_\alpha] \leq \frac{\log(1/\alpha) + O(\log\log(1/\alpha))}{g_P^\star}.$$



Is this test optimal?

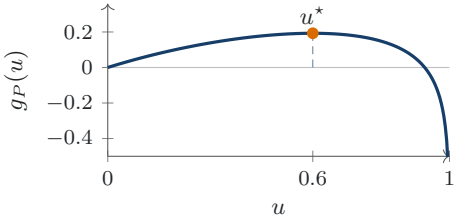
Fundamental limit (Part 1)



Every level-α, power-one test satisfies

$$\mathbb{E}_P[\tau_\alpha] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m})}.$$

Testing-by-betting



The proposed test τ_α achieves

$$\mathbb{E}_P[\tau_\alpha] \leq \frac{\log(1/\alpha)}{g_P^\star}\,(1 + o(1))\,.$$

Is the Kelly growth rate equal to the minimum KL?

$$g_P^\star \stackrel{?}{=} \mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m}).$$

Kelly betting as a minimum-KL dual

For P on $[0,1]$ with $\mathbb{E}_P[X] > m$, recall

$$\mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m}) = \inf_{\substack{Q \in \mathcal{P}([0,1]) \\ \mathbb{E}_Q[X] \le m}} D(P \parallel Q).$$

This is an **infinite-dimensional convex program**, but with one inequality constraint

It turns out that its dual is precisely the **Kelly growth-rate maximization**:

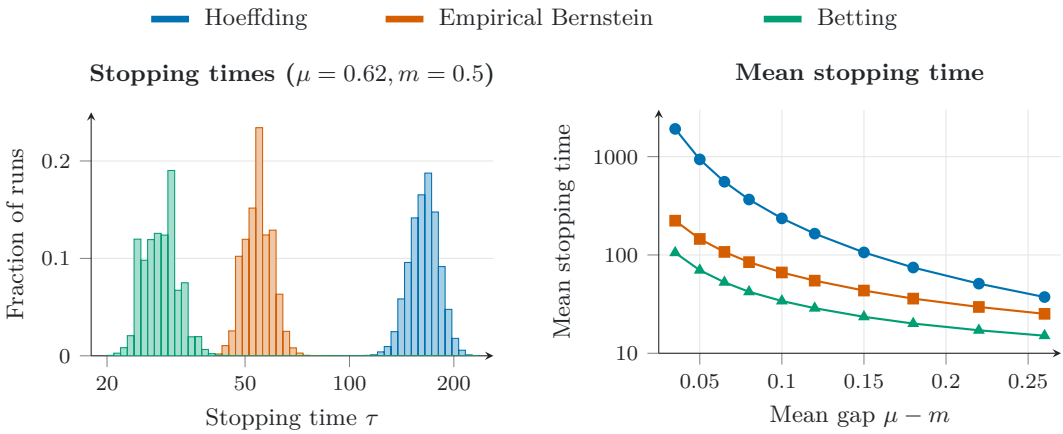
$$\mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m}) = \sup_{u \in [0,1]} \mathbb{E}_P \left[\log \left(1 - u + u \frac{X}{m} \right) \right] := g_P^\star.$$

Combining this duality with our stopping-time bounds,

$$\lim_{\alpha \downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m})}.$$

Empirical Comparison

Baselines: Tests based on time-uniform concentration inequalities

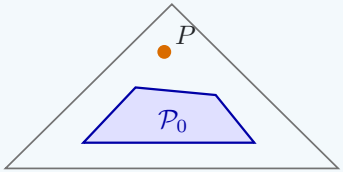


Summary of the two approaches

Finite alphabets

Likelihood-based universal inference

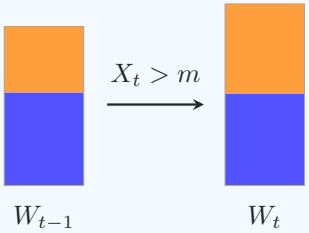
$$E_t = \frac{\text{KT predictor}}{\text{maximum null likelihood}}$$



Bounded-mean testing

Testing by betting

$$W_t = \prod_{i=1}^{t} \left(1 - u_i + u_i \frac{X_i}{m}\right)$$



Is there a common design principle for a general testing problem?

A Unifying Design Principle

Likelihood-free Universal Inference

Testing over general observation spaces

For some Polish space $\mathcal{X}$, let

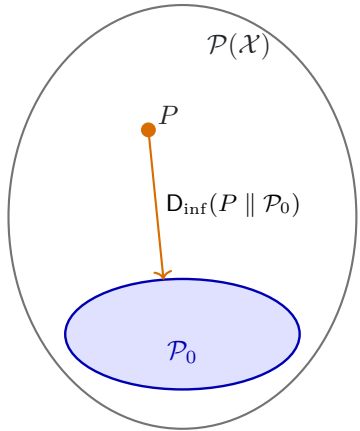
$$X_1, X_2, \ldots \overset{\text{iid}}{\sim} P, \qquad P \in \mathcal{P}(\mathcal{X}) \text{ unknown.}$$

Given $\mathcal{P}_0 \subseteq \mathcal{P}(\mathcal{X})$, we wish to test

$$H_0 : P \in \mathcal{P}_0 \qquad \text{vs.} \qquad H_1 : P \notin \mathcal{P}_0.$$

Lower Bound: For any level-α, power-one test,

$$\mathbb{E}_P[\tau_\alpha] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}, \quad \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) \coloneqq \inf_{Q \in \mathcal{P}_0} D(P\|Q).$$



Can we construct tests attaining this limit without relying on likelihood functions?

Donsker–Varadhan (DV) Representation

For any $P, Q \in \mathcal{P}(\mathcal{X})$, recall the **Donsker–Varadhan variational representation**:

$$\mathsf{D}(P \parallel Q) = \sup_{f \in C_b(\mathcal{X})} \left\{ \mathbb{E}_P[f(X)] - \log \mathbb{E}_Q[e^{f(X)}] \right\},$$

where $C_b(\mathcal{X})$ denotes the space of bounded continuous functions on $\mathcal{X}$.

Interestingly, when $P \ll Q$ and $\log(dP/dQ) \in C_b(\mathcal{X})$, an optimizer is

$$f_Q^\star(x) = \log \frac{dP}{dQ}(x).$$

Key observation: The DV representation allows us to write the log-likelihood ratio as the solution of an optimization problem.

Likelihood-Free UI for $\mathcal{P}_0 = \{Q\}$

For any predictable sequence $f_t \in C_b(\mathcal{X})$, define

$$M_0^Q = 1, \qquad M_t^Q = \prod_{i=1}^{t} \frac{e^{f_i(X_i)}}{\mathbb{E}_Q[e^{f_i(X)}]}.$$

$$X^{t-1} \xrightarrow{\text{Choose}} f_t \xrightarrow{\text{Observe}} X_t \xrightarrow{\text{Update}} M_t^Q$$

Validity. $\{M_t^Q\}$ is a nonnegative martingale under Q:

$$\mathbb{E}_Q[M_t^Q \mid \mathcal{F}_{t-1}] = M_{t-1}^Q \frac{\mathbb{E}_Q[e^{f_t(X)}]}{\mathbb{E}_Q[e^{f_t(X)}]} = M_{t-1}^Q.$$

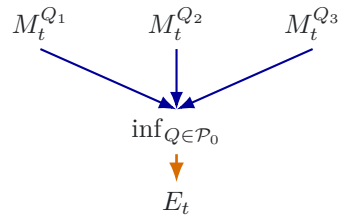
Log-growth. For a fixed function $f_t \equiv f$, under an i.i.d. alternative P,

$$\frac{1}{t} \log M_t^Q \quad = \quad \frac{1}{t} \sum_{i=1}^{t} f_t - \log \mathbb{E}_Q[e^{f_t}] \quad \xrightarrow{\text{a.s.}} \quad \mathbb{E}_P f - \log \mathbb{E}_Q e^f.$$

Likelihood-Free UI for Composite $\mathcal{P}_0$

For each $Q \in \mathcal{P}_0$, construct $\{M_t^Q\}$ using predictable $\{f_{t,Q}\}$

$$M_t^Q = \prod_{i=1}^{t} \frac{e^{f_{i,Q}(X_i)}}{\mathbb{E}_Q[e^{f_{i,Q}(X)}]}.$$



As in Part 1, take the pointwise infimum:

$$E_t = \inf_{Q \in \mathcal{P}_0} M_t^Q.$$

Validity. $\tau_\alpha = \inf\{t \geq 1 : E_t \geq 1/\alpha\}$ satisfies uniform level-α property under $\mathcal{P}_0$.

$E_t \leq M_t^{Q_0}$ Ville's ineq.

$$\sup_{Q_0 \in \mathcal{P}_0} \mathbb{P}_{Q_0}\left(\sup_{t \geq 0} E_t \geq \frac{1}{\alpha}\right) \leq \sup_{Q_0 \in \mathcal{P}_0} \mathbb{P}_{Q_0}\left(\sup_{t \geq 0} M_t^{Q_0} \geq \frac{1}{\alpha}\right) \leq \alpha.$$

Summary of the Likelihood-Free UI Algorithm

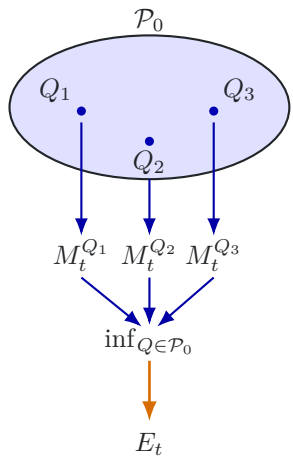
Initialize: $M_0^Q = 1$ for every $Q \in \mathcal{P}_0$.

For $t = 1, 2, \ldots$:

1. **Choose** predictable $f_{t,Q}$ for each $Q \in \mathcal{P}_0$.
2. **Observe** the next X_t.
3. **Update** each null-specific martingale

$$M_t^Q = M_{t-1}^Q \frac{e^{f_{t,Q}(X_t)}}{\mathbb{E}_Q[e^{f_{t,Q}(X)}]}.$$

4. **Aggregate.** $E_t = \inf_{Q \in \mathcal{P}_0} M_t^Q$.
5. **Reject** H_0 if $E_t \geq 1/\alpha$.

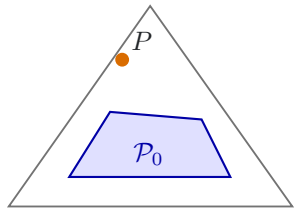


Recovering finite-alphabet universal inference

Suppose $\mathcal{X} = \{1, \ldots, k\}$, and assume each null pmf q is positive.

Predictable function: For each $t \geq 1$, choose

$$f_{t,Q}(x) = \log \frac{\widehat{p}_t(x)}{q(x)}, \quad \text{where} \quad \widehat{p}_t = \text{KT-predictor}$$



Null normalization. For every Q with pmf q, we have

$$\mathbb{E}_Q[e^{f_{t,Q}(X)}] = \sum_{x \in \mathcal{X}} q(x) \frac{\widehat{p}_t(x)}{q(x)} = 1.$$

Substituting into likelihood-free UI gives

$$E_t = \inf_{Q \in \mathcal{P}_0} \prod_{i=1}^{t} \frac{\widehat{p}_i(X_i)}{q(X_i)} = \frac{\widetilde{P}(X^t)}{\sup_{Q \in \mathcal{P}_0} Q^t(X^t)}.$$

Recovering bounded-mean betting

Let $X_t \in [0,1]$ and $\mathcal{P}_0 = \{Q : \mathbb{E}_Q X \le m\}$.

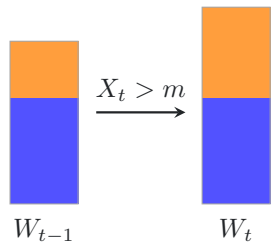
Predictable function: $f_{t,Q} = f_{u_t}$ for predictable u_t, where

$$f_u(x) = \log\left(1 - u + u\frac{x}{m}\right), 0 \le u < 1.$$

Null normalization. For every Q with $\mathbb{E}_Q[X] \le m$,

$$\mathbb{E}_Q[e^{f_{t,Q}(X)}] = 1 - u_t + u_t\frac{\mathbb{E}_Q X}{m} \le 1,$$

with equality if $\mathbb{E}_Q X = m$ (e.g., $Q = \text{Bernoulli}(m)$).



On taking an infimum over Q, we have

$$E_t = \inf_{Q \in \mathcal{P}_0} M_t^Q = \prod_{i=1}^{t}\left(1 - u_i + u_i\frac{X_i}{m}\right) = W_t.$$

Beyond the two examples

Two-sample testing

Observe independent $X_t \sim P_X$ and $Y_t \sim P_Y$.

$$H_0 : \quad P_X = P_Y.$$

Independence testing

Observe i.i.d. pairs $(X_t, Y_t) \sim P_{XY}$.

$$H_0 : \quad P_{XY} = P_X \otimes P_Y.$$

Invariance testing

Observe $X_t \sim P_X$; let G act on $\mathcal{X}$.

$$H_0 : \quad X \overset{d}{=} g(X), \quad \forall g \in G.$$

Log-concavity testing

Observe i.i.d. $X_t \sim P$ on $\mathbb{R}^d$, with density p.

$$H_0 : \quad p \text{ is log-concave.}$$

The same likelihood-free UI idea applies to all these problems.

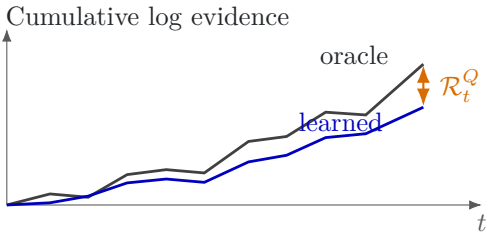
The price of learning: regret

Fix an alternative $P \notin \mathcal{P}_0$. For each $Q \in \mathcal{P}_0$, compare:

- the **oracle** log likelihood process: $\Lambda_t^Q = \sum_{i=1}^t \log \frac{dP}{dQ}(X_i)$,
- the **learned** log evidence $\log M_t^Q$.

Regret: The gap between oracle and learned log evidence,

$$\mathcal{R}_t^Q := \Lambda_t^Q - \log M_t^Q.$$



KT prediction: regret w.r.t. to a fixed distribution.

Universal portfolios: regret w.r.t. to a fixed bet.

Stronger regret control $\Longrightarrow$ stronger guarantees

Fix an alternative P with $\mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0) = \inf_{Q \in \mathcal{P}_0} \mathsf{D}(P \,\|\, Q) > 0$.

Let $\mathcal{R}_t = \sup_{Q \in \mathcal{P}_0} \mathcal{R}_t^Q$ be the *uniform regret* over the null distribution class.

Control on $\mathcal{R}_t$	**Statistical Guarantee**	
$\limsup_{t \to \infty} \dfrac{\mathcal{R}_t}{t} < \mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)$ *a.s.*	$\mathbb{P}_P(\tau_\alpha < \infty) = 1$	(Power one)
$\mathcal{R}_t \le \rho t$, $\rho < \mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)$, *w.h.p.*	$\mathbb{E}_P[\tau_\alpha] < \infty$	(Finite expectation)
$\mathcal{R}_t \le r_t$ with $r_t = o(t)$, *w.h.p.*	$\mathbb{E}_P[\tau_\alpha] \sim \dfrac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)}$	(First-order opt.)

Sufficient conditions for power one property

Under certain conditions, we can show that

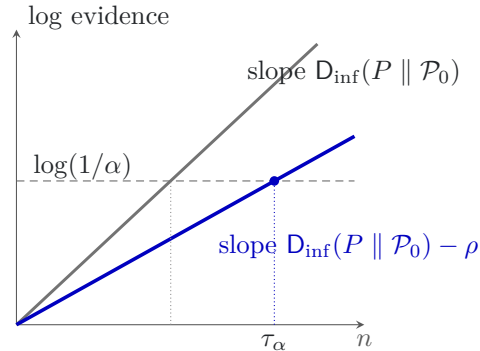
$$\frac{\log E_t}{t} \gtrsim \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) - \frac{\mathcal{R}_t}{t}.$$

Suppose $\limsup_{t\to\infty} \frac{\mathcal{R}_t}{t} \leq \rho < \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)$ a.s.

Then, the asymptotic growth rate is positive:

$$\liminf_{t\to\infty} \frac{\log E_t}{t} \geq \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) - \rho > 0.$$

Hence, $\{E_t : t \geq 1\}$ crosses $\log(1/\alpha)$ w.p. 1.

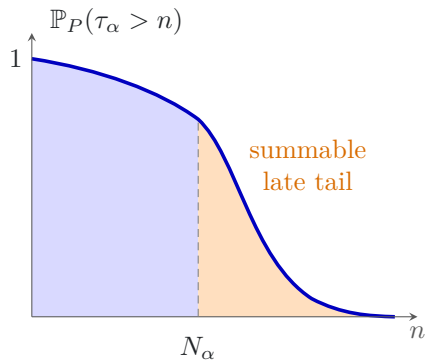


From regret to expected stopping time

$$\mathbb{E}_P[\tau_\alpha] = \sum_{n\geq 0} \mathbb{P}_P(\tau_\alpha > n) \leq N_\alpha + \underbrace{\sum_{n\geq N_\alpha} \mathbb{P}_P(\tau_\alpha > n)}_{\text{Late-stopping tail}}.$$

Finite mean: A summable late-stopping tail for some N_α is sufficient.

Optimality: Choose $N_\alpha = \left\lceil \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P\|\mathcal{P}_0)-\varepsilon} \right\rceil$.



Under stronger regret control, can show

$$\mathbb{E}_P[\tau_\alpha] \leq N_\alpha + O_{P,\varepsilon}(1).$$

Letting $\alpha \downarrow 0$ and then $\varepsilon \downarrow 0$:

$$\lim_{\alpha\downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}.$$

Summary of the general design principle

1. **Construct null-specific martingales.**
   For every $Q \in \mathcal{P}_0$, construct a nonnegative martingale $\{M_t^Q\}_{t\geq 0}$ under Q, using predictable witness functions $\{f_{t,Q} : t \geq 1\}$.

2. **Aggregate by pointwise minimization.**

$$E_t = \inf_{Q\in\mathcal{P}_0} M_t^Q.$$

   The resulting e-process is anytime-valid under every $Q \in \mathcal{P}_0$.

3. **Control regret to guarantee efficiency.**
   Stronger regret and tail guarantees yield stronger statistical performance:

$$\text{Power one} \quad \longrightarrow \quad \text{Finite } \mathbb{E}_P[\tau_\alpha] \quad \longrightarrow \quad \text{First-order optimality.}$$

Some Extensions Beyond I.I.D. Testing

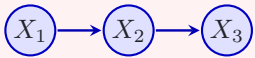
Beyond the basic i.i.d. testing problem

Our basic i.i.d. testing framework has three ingredients:

Observation model	**Inferential task**	**Performance criterion**
$X_t \overset{\text{i.i.d.}}{\sim} P$	$H_0 : \ P \in \mathcal{P}_0$	$\mathbb{E}_P[\tau_\alpha]$
Fully observed, fixed P	Reject or continue	Stop quickly on average

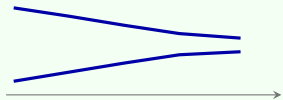
Change the observations

Markov chains; censored data



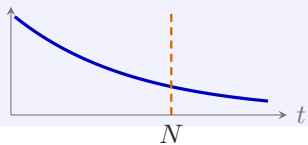
Change the task

Confidence sequences; change detection



Change the objective

Deadlines; time-sensitive rewards



Modified Task 1: Estimation

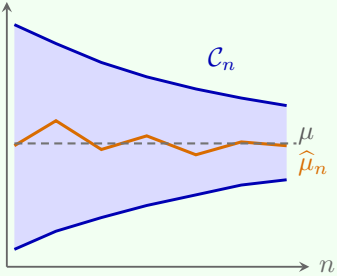
Suppose $X_1, X_2, \ldots \overset{\text{i.i.d.}}{\sim} P$, and we want to estimate the mean $\mu = \mathbb{E}_P[X]$.

Formally, this involves constructing intervals $\{\mathcal{C}_n\}_{n \geq 1}$, adapted to the data, such that

$$\mathbb{P}_P(\forall n \geq 1 : \ \mu \in \mathcal{C}_n) \geq 1 - \alpha.$$

This is a level-$(1-\alpha)$ confidence sequence (CS) for μ

Equivalently, $\mathbb{P}_P(\exists n \geq 1 : \ \mu \notin \mathcal{C}_n) \leq \alpha$



Contrast with fixed-time confidence intervals:

$$\underbrace{\mathbb{P}_P(\mu \in \mathcal{C}_n) \geq 1 - \alpha}_{\text{fixed } n} \qquad \text{vs.} \qquad \underbrace{\mathbb{P}_P(\forall n \geq 1 : \ \mu \in \mathcal{C}_n) \geq 1 - \alpha}_{\text{all times simultaneously}}.$$

The coverage guarantee holds for all times simultaneously; hence for all random stopping times as well.

Constructing CSs by inverting anytime-valid tests

For each candidate mean $m \in [0,1]$, test the point null $H_{0,m}: \ \mu = m$ using an evidence process $\{M_n(m)\}_{n\geq 1}$.

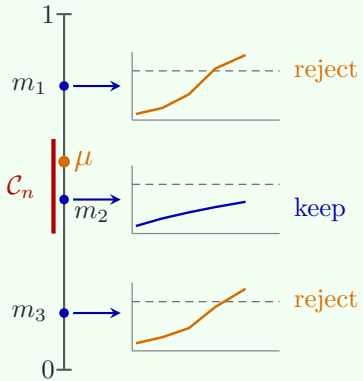
Invert the tests. Keep exactly those values that have not yet been rejected:

$$\mathcal{C}_n = \left\{ m : \ \max_{1\leq s\leq n} M_s(m) < \frac{1}{\alpha} \right\}.$$

At $m^\star = \mu$, we have by Ville's inequality,

$$\mathbb{P}_P(\exists n \geq 1 : \ m^\star \notin \mathcal{C}_n) = \mathbb{P}_P\left(\sup_{n\geq 1} M_n(m^\star) \geq \frac{1}{\alpha} \right) \leq \alpha.$$

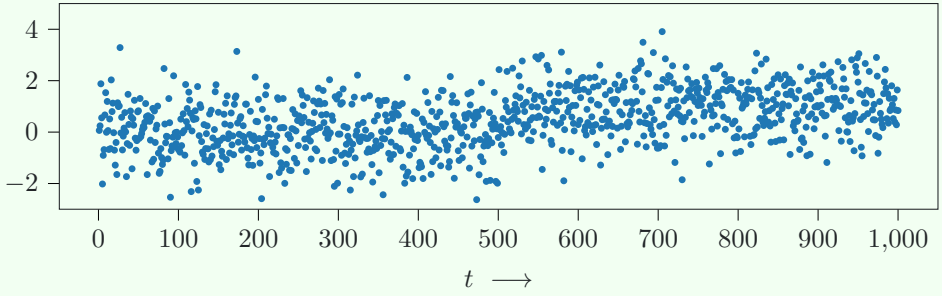
We can also characterize the optimal size of $|\mathcal{C}_n|$ in terms of inverse $\mathsf{D}_{\mathsf{inf}}$ terms.



Modified Task 2: Sequential change detection

Suppose the distribution of the observations changes at an unknown time ν:

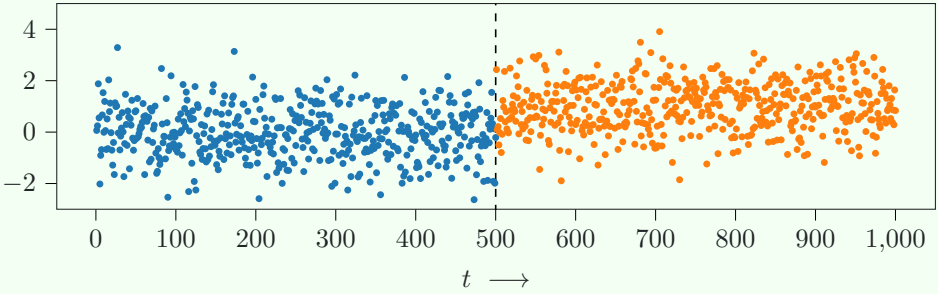
$$X_t \sim \begin{cases} P_{\text{pre}}, & t \leq \nu, \\ P_{\text{post}} \neq P_{\text{pre}}, & t > \nu. \end{cases}$$



Modified Task 2: Sequential change detection

Suppose the distribution of the observations changes at an unknown time ν:

$$X_t \sim \begin{cases} P_{\text{pre}}, & t \le \nu, \\ P_{\text{post}} \neq P_{\text{pre}}, & t > \nu. \end{cases}$$

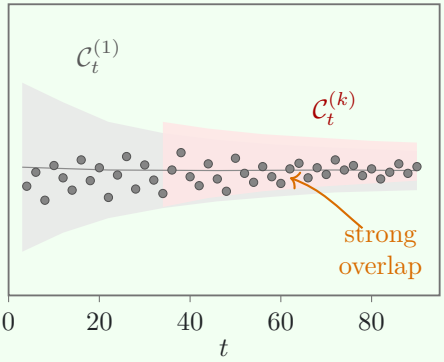


Goal: Detect the change quickly while keeping false alarms rare.

Change detection using confidence sequences

1. At each round k, start a new CS $\{\mathcal{C}_t^{(k)} : t \geq k\}$.
2. At time t, compare all active confidence sequences $\mathcal{C}_t^{(1)}, \mathcal{C}_t^{(2)}, \ldots, \mathcal{C}_t^{(t)}$.
3. Stop as soon as they *disagree*, i.e.

$$\tau = \inf\Big\{t \geq 1 : \bigcap_{k=1}^{t} \mathcal{C}_t^{(k)} = \varnothing\Big\}.$$

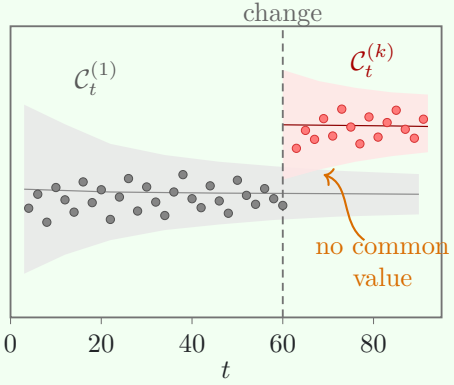


- When $\nu = \infty$, we have $\mathbb{E}_\infty[\tau] \geq \frac{1}{\alpha}$.
- When $\nu < \infty$, detection delay depends on quality of the CSs used.

Change detection using confidence sequences

1. At each round k, start a new CS $\{\mathcal{C}_t^{(k)} : t \geq k\}$.
2. At time t, compare all active confidence sequences $\mathcal{C}_t^{(1)}, \mathcal{C}_t^{(2)}, \ldots, \mathcal{C}_t^{(t)}$.
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- When $\nu = \infty$, we have $\mathbb{E}_\infty[\tau] \geq \frac{1}{\alpha}$.
- When $\nu < \infty$, detection delay depends on quality of the CSs used.

Modified Observation Model 1: Markov chains

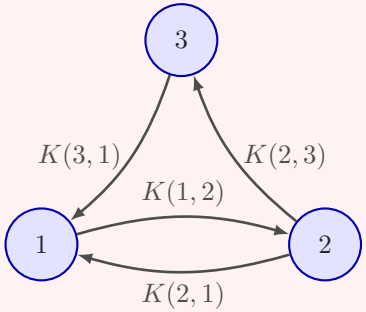
Consider a Markov chain on a finite state space $\mathcal{S} = \{1, \ldots, k\}$.

Let $K(\cdot, \cdot)$ denote the **unknown** transition matrix:

$$\mathbb{P}_K(X_t = y \mid \mathcal{F}_{t-1}) = K(X_{t-1}, y).$$

Testing problem

$$H_0 : \ K \in \mathcal{K}_0 \qquad \text{vs.} \qquad H_1 : \ K \notin \mathcal{K}_0.$$



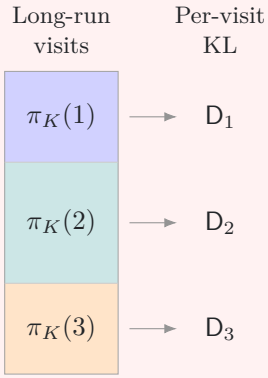
Goal: Design a level-α power-one test τ for the above problem with small $\mathbb{E}_K[\tau]$.

The lower bound for Markov chains

Consider an ergodic alternative K with stationary distribution π_K

The relevant information quantity is

$$\mathsf{D}_{\inf}^{\mathrm{MC}}(K \parallel \mathcal{K}_0) := \inf_{L \in \mathcal{K}_0} \sum_{x \in \mathcal{S}} \pi_K(x)\, \mathsf{D}\big(K(x,\cdot) \parallel L(x,\cdot)\big).$$



Lower bound. Every level-α, power-one test satisfies

$$\mathbb{E}_K[\tau_\alpha] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}^{\mathrm{MC}}(K \parallel \mathcal{K}_0)} - O_K(1).$$

Proof: Combination of DPI and a modified Wald's identity.

Universal inference for Markov Chains

Finite-state Markov UI: Replace symbol pmfs in the i.i.d. case by conditional transition probabilities

$$M_t^{\mathrm{MC}} = \frac{\prod_{i=2}^{t} \widehat{K}_i(X_i \mid X_{i-1})}{\sup_{L \in \mathcal{K}_0} \prod_{i=2}^{t} L(X_{i-1}, X_i)} = \frac{\text{Mixture of Conditional Likelihoods}}{\text{Maximum Null Conditional Likelihood}}.$$

Stopping time: $\tau_\alpha = \inf\{t \geq 2 : M_t^{\mathrm{MC}} \geq 1/\alpha\}$

Unchanged: Nullwise domination + Ville still guarantee anytime validity.

New tools: Universal predictors for Markov sources; Markov concentration; modified Wald's identity.

Under suitable conditions, $\mathbb{E}_K[\tau_\alpha] \sim \log(1/\alpha)/\mathsf{D}_{\inf}^{\mathrm{MC}}(K \parallel \mathcal{K}_0)$.

Modified objective 1: Horizon-aware testing

So far, we have focused on minimizing $\mathbb{E}_P[\tau_\alpha]$.

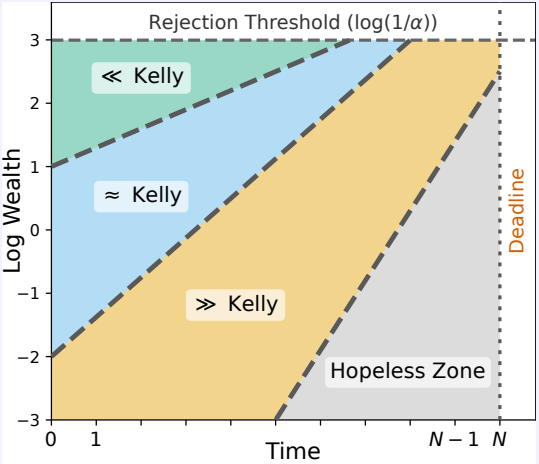
What if we have a **strict deadline** N?

Horizon-aware objective.

$$\begin{aligned} &\max \mathbb{P}_P(\tau_\alpha \le N), \\ \text{s.t.} \quad &\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau_\alpha \le N) \le \alpha. \end{aligned}$$

For the betting test, it reduces to

$$\max_{\{u_t\}_{t=1}^N} \mathbb{P}_P\left(\max_{1 \le t \le N} \prod_{i=1}^{t} (1 - u_i + u_i \frac{X_i}{m}) \ge \frac{1}{\alpha} \right).$$



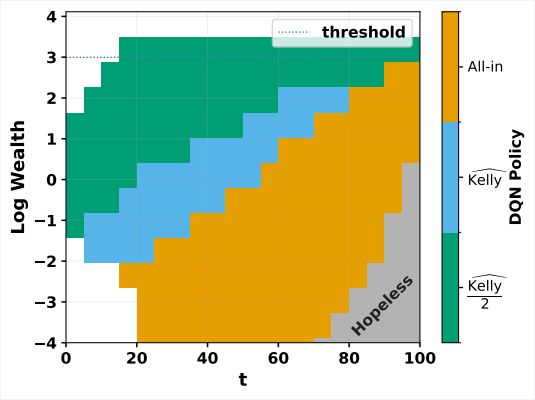
Horizon-aware betting

Finite-horizon control. For a fixed alternative P, the state is $(t, \log W_t)$; the action is the next feasible bet.

Theory. Under suitable conditions:

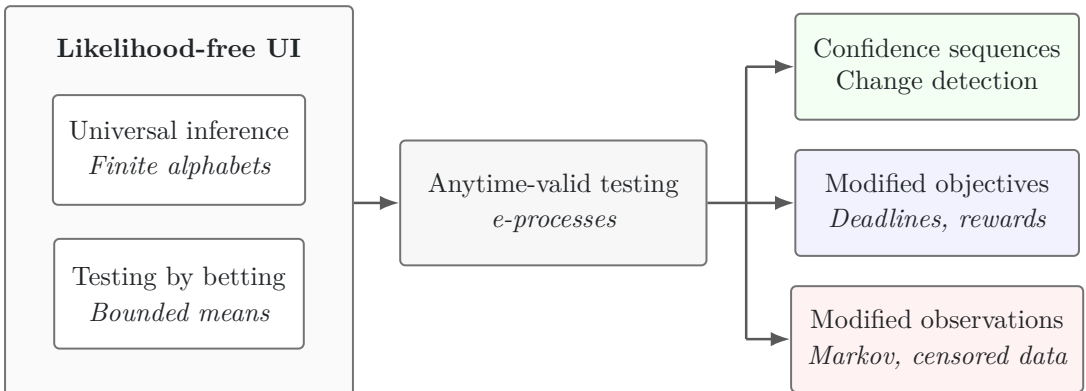
- **Ahead:** less aggressive than Kelly.
- **On schedule:** approximately Kelly.
- **Behind:** more aggressive than Kelly.

Unknown P. A Deep Q-Network (DQN) can learn to select bets using statistics of past observations.



Learned DQN actions.

Summary so far



Some Future Directions

Beyond first-order optimality

Our benchmark: for every fixed alternative P,

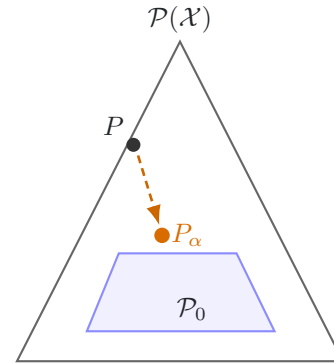
$$\mathbb{E}_P[\tau_\alpha] \sim \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)} \qquad (\alpha \downarrow 0).$$

Second-order behavior.

How does the stopping time fluctuate around its leading term?

Time-uniform local detection boundary.

What if the alternative P approaches $\mathcal{P}_0$ as $\alpha \downarrow 0$?



Active anytime-valid inference

Multiple data sources. Suppose K sources are governed by a common hypothesis.

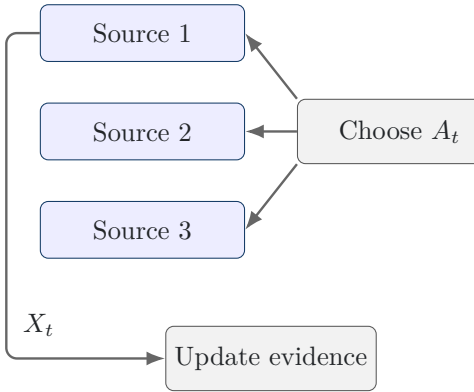
At each time t, the analyst chooses

$$A_t \in \{1, \ldots, K\}, \qquad X_t \sim P_{A_t},$$

using the observations collected so far.

Different sources may provide different amounts of information.

Question. What is the optimal joint policy for *sensing and stopping*?

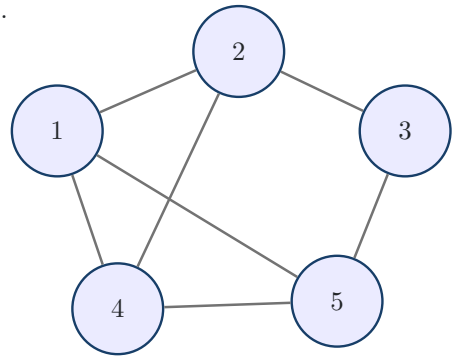


Distributed anytime-valid inference

Distributed observations: multiple agents collect local data, satisfying a common underlying hypothesis.

Each agent maintains local evidence $E_t^{(i)}$ and communicates with its neighbors (or a central server).

Question: How can a network aggregate evidence with both validity and optimal information growth?



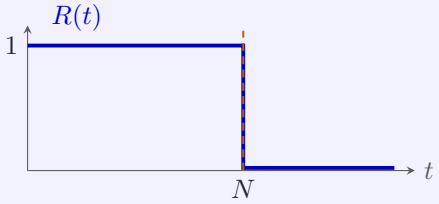
Thank you.

Modified objective 2: Time-sensitive testing

A deadline is one way to reward early detection. More generally, assign a nonnegative, nonincreasing reward $R(t)$ to rejection at time t.

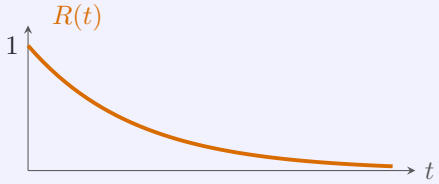
$$\sup_{\tau_\alpha \in \mathcal{T}_\alpha} \mathbb{E}_P[R(\tau_\alpha)], \qquad R(\infty) = 0,$$

Hard deadline



$R(t) = \mathbf{1}\{t \le N\}$ gives $\mathbb{P}_P(\tau_\alpha \le N)$.

Discounted discovery



$R(t) = e^{-ct}$ gives $\mathbb{E}_P[e^{-c\tau_\alpha}]$.

Rényi divergences and reward decay

Converse (ongoing work). For a finite-alphabet alternative $P \notin \mathcal{P}_0$, let $\mathcal{I}_r(P) := \inf_{Q \in \mathcal{P}_0} D_r(P \,\|\, Q)$.
Write $\Phi_P(\eta) := \eta \mathcal{I}_{1/(1-\eta)}(P)$, where $0 < \eta < 1$. Every level-α test satisfies

$$\mathbb{E}_P[R(\tau_\alpha)] \;\le\; \inf_{0<\eta<1} \left\{ \alpha^\eta \sup_{n \ge 1} R(n) e^{n\Phi_P(\eta)} \right\}.$$

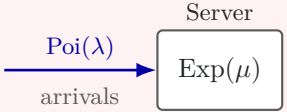
Three regimes, depending on how rapidly the reward decays:

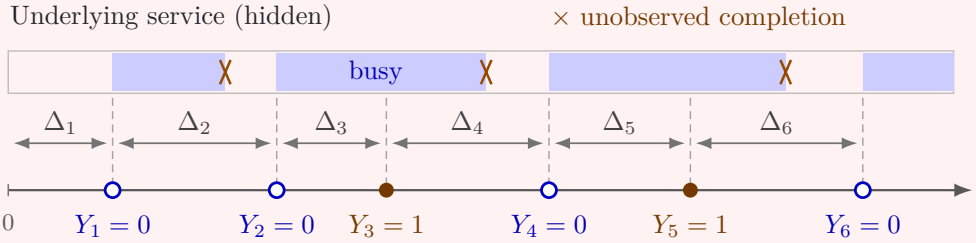
Subexponential $R(n) = e^{-o(n)}$	KL / typical stopping time $\log(1/\alpha)/\mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)$; a separate argument is needed.
Exponential $R(n) = e^{-cn}$	For $0 < c < \mathcal{I}_\infty(P)$, the Rényi exponent η_c solves $\Phi_P(\eta_c) = c$.
Superexponential $R(n) \lesssim e^{-cn}$	If $c \ge \mathcal{I}_\infty(P)$, the reward is $O(\alpha)$; randomized early rejection attains this order.

Here $\mathcal{I}_\infty(P) := \inf_{Q \in \mathcal{P}_0} D_\infty(P \,\|\, Q)$.

Modified Observations 2: Censoring

- A single server with service times $S \sim \text{Exp}(\mu)$.
- Jobs arrive as a Poisson process (rate λ).
- An arriving job is admitted whenever the server is idle.





Observations: inter-arrival times $\{\Delta_t : t \geq 1\}$ and busy/idle indicators $\{Y_t : t \geq 1\}$.

Goal: For some known $\mu^\star > 0$, decide between $H_0 : \ \mu \geq \mu^\star$ vs. $H_1 : \ \mu < \mu^\star$.

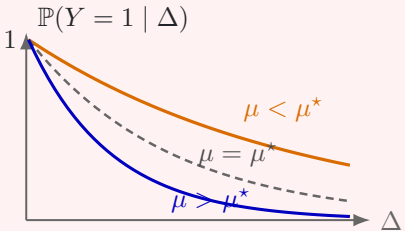
Anytime-Valid Test From Occupancy Indicators

Challenge: We want to test a property of service times $\{S_t : t \geq 1\}$, but we don't directly observe them!

Under the Poisson-arrival assumption: $\Delta_t \sim \mathrm{Exp}(\lambda)$.

Under the additional Exponential service-time assumption,

$$Y_t \mid (\Delta_t, \mathcal{F}_{t-1}) \sim \mathrm{Ber}(e^{-\mu \Delta_t}).$$



An instantiation of the likelihood-based universal inference procedure gives, for fixed $\mu < \mu^\star$,

$$\mathbb{E}_{\mu,\lambda}[\tau_\alpha] \sim \frac{\log(1/\alpha)}{I_{\mu,\lambda}}, \quad \text{where} \quad I_{\mu,\lambda} = \mathbb{E}_\Delta\left[\mathsf{d}\big(e^{-\mu\Delta}, e^{-\mu^\star \Delta}\big)\right],$$

Similar ideas extend to more general service time distributions.